MAE 145: Planning and Estimation
Homework #5

Assigned May 1. Due on Friday, May 15

Notice:

Please upload 3 files for this homework. One to include the written part of the homework (problem 1, problem 2, and parts (i) and (ii) of problem 3), a python file for the second exercise using the template `python_program_template_hw5.py`, and the `main.py` if you choose do to the last problem. In your assignment include the names of the other students you have collaborated with to do the homework problems. The Collaboration Policy for this course is detailed in the syllabus.

Homework exercises:

1. E4.2 from the Lecture Notes on Robotics Planning and Kinematics.
2. E5.2 from the Lecture Notes on Robotics Planning and Kinematics.
3. E4.3 from the Lecture Notes on Robotics Planning and Kinematics.

   Programming guidelines:

   The name of your python file should be python_program.py with the first line in the file having your PID as a comment, Ex: # A432443.

   The function names must be adhered to as mentioned in the problem statement. All the assumptions in the problem such as $n = k^2$ for a Sukharev grid and the two prime numbers for a Halton series will be accounted for when testing the code i.e. you will not have to check for prime numbers or perfect squares of your inputs yourself, but the program will give you always a message.

Example:
$X, Y = \texttt{computeGridSukharev}(n)$
$X, Y = \texttt{computeGridSukharev}(4)$
$X = [0.25, 0.25, 0.75, 0.75]$
$Y = [0.25, 0.75, 0.25, 0.75]$
The points are $(X[i], Y[i]) : (0.25, 0.25), (0.25, 0.75), (0.75, 0.25), (0.75, 0.75)$
Similarly,
$X, Y = \texttt{computeGridRandom}(n)$
$X, Y = \texttt{computeGridRandom}(4)$
$X, Y = \texttt{computeGridHalton}(n, b1, b2)$
$X, Y = \texttt{computeGridHalton}(4, 2, 3)$
The order of the points in the list X and Y do not matter

Note: For this assignment you will be evaluated by plotting the sampled points in a unit square as shown below. For you to plot in python you can use a library called matplotlib. The plotting functions in this library is very similar to MATLAB. Ensure to set a limit for the x an y axis from 0 to 1.

Example:

```
import matplotlib.pyplot as plt
plt.scatter(X, Y, color='r', marker='o')
plt.title("Sukharev Grid")
plt.xlim((0,1))
plt.ylim((0,1))
plt.show()
```

4. **(Optional)** Extra credit programming exercise. *Note:* This problem requires familiarity with python classes and object oriented programming. If you are not sufficiently familiar with it, you can skip this problem. The TA will explain main parts of it in the discussion section.

   In this exercise you have to program an RRT planner. To do this, retrieve the files in the folder scr-student available in canvas. In this folder you will find instructions (see Readme.md), a file `main.py` and `auxiliary_functions.py` (halton point in segment computation) that you have to edit, and some auxiliary file containing some functions that you can use for this. You will also see two figures: rrt.png is what you should obtain, while the rrt-student.png is what the program would output without any modifications to it. You have to do 4 tasks:

   (a) Find the nearest node in the tree to the sampled point

   (b) Find the distance of the random sampled point to the nearest node in the tree and if the distance is greater than 0.25; you can use that point to build your tree. If the sampled point is further away, then create a new point, `q-new` $= 0.75 \cdot$ `q-expansion` $+ 0.25 \cdot$ `q-random`, where `q-expansion` is the nearest node in the tree and `q-random` is the random sample point.

   (c) Check if the new point connects to the tree you are building without hitting an obstacle. You can use the auxiliary function to do this. You might want to tweak the parameters to sample the line segment.

   (d) If you can connect the tree to the new point by a collision-free path, make this point a node in the tree with its parent as the nearest node to the tree.

   In addition, tune the choice $(0.75, 0.25)$ in part (b) to other values of (α_1, α_2) such that $\alpha_1 + \alpha_2 = 1$, and $\alpha_1, \alpha_2 \in [0, 1]$, to see how this affects your results.